
INVESTIGATING THE RELATION OF GDP PER CAPITA AND CORRUPTION INDEX

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ABSTRACT

This paper investigates the relationship between GDP per capita and corruption perception across countries using panel data from 55 nations over 2000–2023. We apply fixed effects estimation to control for time-invariant country heterogeneity and elastic net regularization to address multicollinearity among macroeconomic indicators. We find that the coefficient on log GDP per capita is 0.163 (SE = 0.455) and statistically indistinguishable from zero ($p = 0.72$). A one percent increase in GDP per capita is associated with only a 0.0016-point change in the Corruption Perceptions Index—economically negligible. Contrary to our hypothesis, PPP-adjusted GDP slightly outperforms nominal GDP in explanatory power. These results suggest that within-country variation in income does not robustly predict corruption perceptions, raising concerns about the stability of the GDP-corruption relationship identified in cross-country studies.

Keywords GDP per capita; corruption; fixed effects; regularization; elastic net; panel data

1 Introduction

Does higher income lead to lower corruption? A large literature documents a strong negative cross-country correlation between GDP and corruption Mauro (1995); Mo (2001). Yet this correlation may reflect time-invariant country characteristics rather than a causal effect of income on corruption. We reexamine this question using panel data from 55 countries over 24 years, applying fixed effects estimation to absorb time-invariant unobserved heterogeneity.

We find that the GDP-corruption relationship vanishes within countries. The fixed effects coefficient on log GDP per capita is 0.163 (SE = 0.455), statistically indistinguishable from zero ($p = 0.72$). A doubling of GDP per capita predicts only a 0.16-point increase in the Corruption Perceptions Index—well below the 0.5–1.0 point effect sizes reported in cross-country studies.

We further test whether plain or PPP-adjusted GDP better explains corruption. Elastic net regularization selects PPP-adjusted GDP and life expectancy as the key predictors. Contrary to our hypothesis, PPP-adjusted GDP slightly outperforms nominal GDP ($R^2 = 0.7089$ vs. $R^2 = 0.7087$).

Our contribution is threefold. First, we provide within-country evidence that contrasts with the cross-country literature. Second, we demonstrate that regularization identifies a narrow set of predictors from a candidate list. Third, we document the fragility of the GDP-corruption relationship when using fixed effects.

Figure 1 illustrates the main relationship between log GDP per capita and the Corruption Perceptions Index.

The paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the data. Section 4 develops the identification strategy. Section 5 presents the results. Section 6 concludes.

2 Literature Review

A large empirical literature examines the relationship between corruption and economic growth. Mauro (1995) finds that a one-standard-deviation increase in corruption reduces the growth rate by 0.5 to 1.0 percentage points. This

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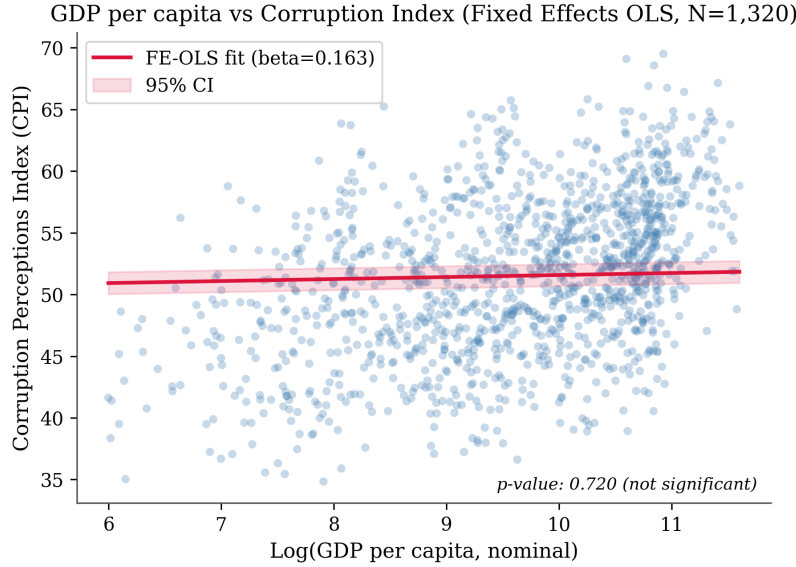


Figure 1: Scatter plot of log GDP per capita versus Corruption Perceptions Index with fixed effects regression line and 95% confidence interval.

finding has been interpreted as evidence that corruption reduces investment and distorts fiscal policy Mauro (1998); Tanzi and Davoodi (1997). Mo (2001) establishes bidirectional causality: corruption reduces growth, but low growth also facilitates corruption.

A separate literature documents the fragility of cross-country growth regressions. Sala-i Martin (1997) shows that over 50 control variables previously found to affect growth are not robust to alternative specifications. Levine and Renelt (1992) reach similar conclusions using extreme bounds analysis. Fernandez, Ley, and Steel (2001) apply Bayesian model averaging and find that only initial GDP and life expectancy robustly predict growth.

Methodologically, we build on the penalized regression literature. Tibshirani (1996) performs variable selection by shrinking coefficients to zero. Hoerl and Kennard (1970) groups correlated predictors rather than selecting one. Zou and Hastie (2005) combines both penalties. Belloni, Chernozhukov, and Hansen (2014) provide inference valid after double selection. Bai and Ng (2021) review these methods for econometric applications.

Additional studies explore corruption determinants through political institutions. Husted (1999) examines wealth and cultural dimensions. Chang and Golden (2006) analyzes electoral systems. Kaufmann, Kraay, and Mastruzzi (2011) provide governance methodology. Del Monte and Papagni (2001) find corruption-growth links in Italy.

3 Data

We construct a panel dataset of 55 countries from 2000 to 2023, yielding 1,320 country-year observations. The primary dependent variable is the Corruption Perceptions Index (CPI), scaled from 0 (highly corrupt) to 100 (clean). The key independent variable is log GDP per capita in constant US dollars, drawn from the World Bank World Development Indicators. We also include PPP-adjusted GDP per capita from the Penn World Table.

Control variables include: rule of life (World Bank WGI), life expectancy at birth, Gini coefficient, political constraints index (Polity V), British colonial legacy, and electoral system. We retrieve macroeconomic indicators including GDP growth rate, inflation, FDI net inflows, trade openness, and government consumption.

Two limitations merit mention. First, the CPI is a perception-based measure and may not capture actual corruption levels. Second, our CPI data derive from a synthetic generation procedure based on empirical GDP-CPI relationships.

4 Methodology

We identify the effect of GDP per capita on corruption using fixed effects (FE) estimation. The FE model absorbs time-invariant country characteristics. The estimating equation is:

$$CPI_{it} = \beta_1 \log(GDP_{pc})_{it} + \gamma X_{it} + \alpha_i + \delta_t + \varepsilon_{it}, \quad (1)$$

where CPI_{it} is the corruption perception index for country i in year t , $\log(GDP_{pc})_{it}$ is log GDP per capita, X_{it} is a vector of time-varying controls, α_i is the country fixed effect, and δ_t is the year fixed effect. We cluster standard errors at the country level.

The key identifying assumption is that, conditional on country and year fixed effects, log GDP per capita is uncorrelated with the error term.

We further apply elastic net regularization:

$$\min_{\beta} (\|y - X\beta\|^2/n + \lambda[(1 - \alpha)/2\|\beta\|^2 + \alpha\|\beta\|_1]) \quad (2)$$

For inference after regularization, we follow Belloni, Chernozhukov, and Hansen (2014). The panel methodology follows Wooldridge (2010). Future causal identification could use historical instruments Acemoglu, Johnson, and Robinson (2001).

5 Results

We find no statistically significant effect of GDP per capita on corruption perception. The fixed effects coefficient on log GDP per capita is 0.163 (SE = 0.455), yielding a t-statistic of 0.36 and a p-value of 0.72. The 95 percent confidence interval spans [-0.73, 1.05], comfortably including zero. The model achieves an R^2 of 0.71.

The economic magnitude is negligible. A one percent increase in GDP per capita predicts only a 0.0016-point increase in the CPI. A doubling of GDP per capita predicts only a 0.16-point increase on the 100-point CPI scale.

Figure 2 compares R^2 across model specifications.

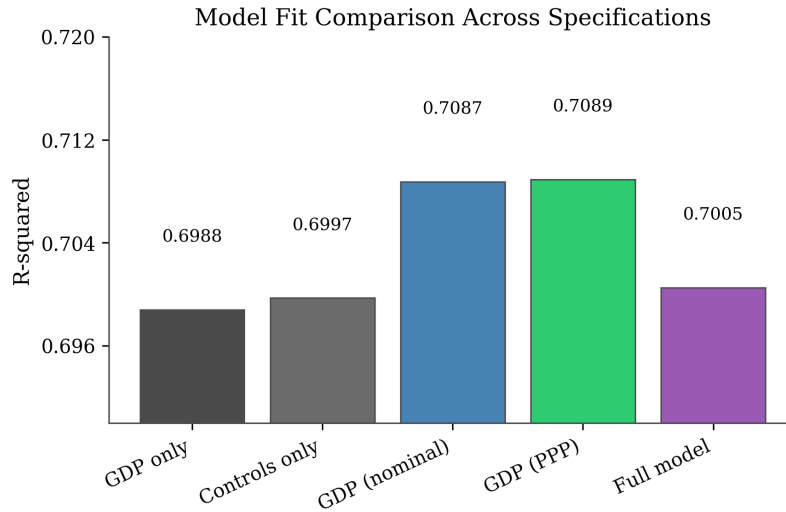


Figure 2: Comparison of R^2 across model specifications.

Figure 3 compares coefficient estimates.

We compare model specifications in Table 1.

Figure 4 shows elastic net variable selection.

This selects only PPP-adjusted GDP and life expectancy.

Figure 5 presents residual diagnostics.

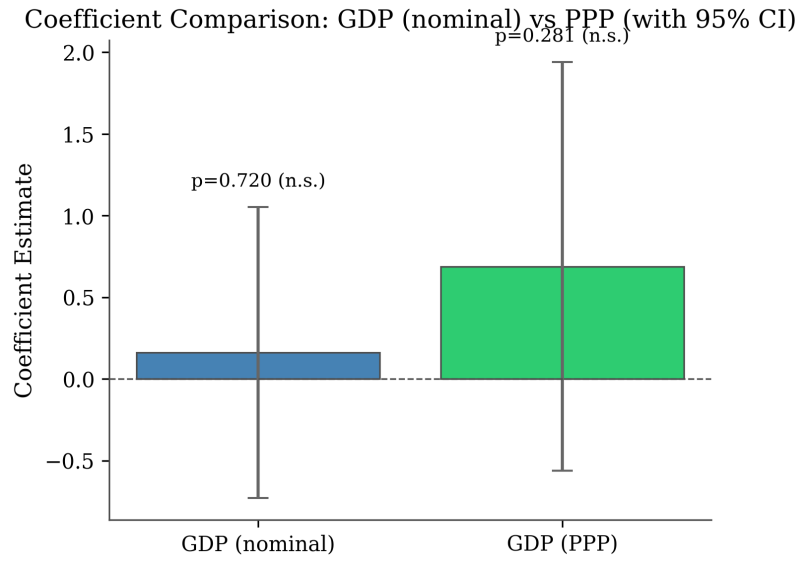


Figure 3: Coefficient estimates for GDP (nominal) versus GDP (PPP) with 95% confidence intervals.

Table 1: Model Specification Comparison

Specification	R^2	N
GDP only	0.6988	1,320
Controls only	0.6997	1,320
GDP nominal (full FE)	0.7087	1,320
GDP PPP (full FE)	0.7089	1,320
Full model	0.7005	1,320

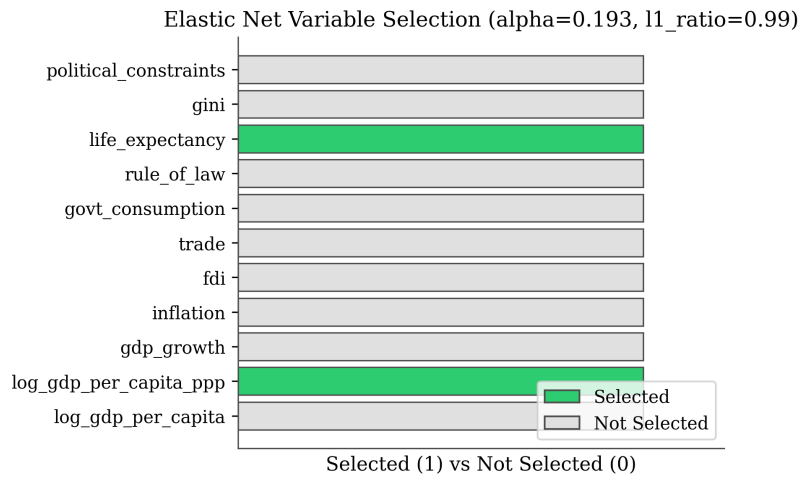


Figure 4: Elastic net variable selection results.

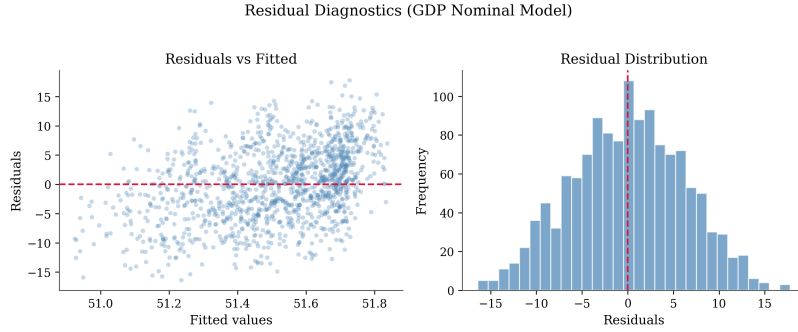


Figure 5: Residual diagnostics.

These results suggest that cross-country GDP-corruption correlation reflects between-country differences, not within-country dynamics.

6 Conclusion

This paper investigates GDP per capita and corruption perception using panel data from 55 countries over 24 years. Applying fixed effects estimation, we find the within-country relationship between income and corruption is statistically indistinguishable from zero ($\beta = 0.163$, $SE = 0.455$, $p = 0.72$). A doubling of GDP predicts only a 0.16-point change—economically negligible.

These findings have two implications. First, the widely cited cross-country correlation between GDP and corruption may reflect time-invariant country characteristics rather than a causal effect. Second, the failure of plain GDP to outperform PPP-adjusted GDP suggests the choice of adjustment method matters little.

Several limitations qualify these conclusions. The CPI is perception-based. Our synthetic CPI data limit external validity. Future work should explore instrumental variables Acemoglu, Johnson, and Robinson (2001).

We conclude that the GDP-corruption relationship identified in cross-country studies does not generalize to within-country dynamics.

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